

AI 2002 – Artificial Intelligence (Spring 2025)

Assignment 4

Topics Covered: FOL and Clustering	Submission Deadline: Tuesday – May 06, 2025 by 23.00 sharp
Submission Guidelines: Group assignment: 2 members (at max.) or an individual - Submit a PDF file containing your hand-written answers to Question 1 & 2. - Submit. ipynb files for Question 3 & 4 that contain your code and textual answers where required. Alternatively, you may use a PDF file to submit your answers for analysis tasks.	

Question 1: FOL Basics

For each pair of atomic sentences, give the most general unifier if it exists:

1. $P(A,B,B), P(x,y,z)$.
2. $Q(y,G(A,B)), Q(G(x,x),y)$.
3. $Older(Father(y),y), Older(Father(x),John)$.
4. $Knows(Father(y),y), Knows(x,x)$.

Question 2: FOL Inference

1. Consider the following knowledge base in FOL.

Knowledge Base (KB):

1. All professors who teach a course are experienced.
 $\forall x, y: Teaches(x, y) \rightarrow Experienced(x)$
2. Dr. Smith teaches AI.
 $Teaches(Smith, AI)$
3. A person is qualified to supervise thesis if they are experienced and have taught AI.
 $\forall x: (Experienced(x) \wedge Teaches(x, AI)) \rightarrow Qualified(x)$

4. Only qualified people can supervise students.

$\forall x: \text{Qualified}(x) \rightarrow \text{CanSupervise}(x)$

Using resolution, prove whether the following statement logically follows from the KB: Can Dr. Smith supervise students? (formally, $\text{CanSupervise}(\text{Smith})$)

2. Suppose a knowledge base contains just the following first-order (Horn) clauses.

- $\text{Ancestor}(\text{Mother}(x), x)$
- $\text{Ancestor}(x, y) \wedge \text{Ancestor}(y, z) \Rightarrow \text{Ancestor}(x, z)$

Can a resolution algorithm prove the sentence $\neg \text{Ancestor}(\text{John}, \text{John})$ from the original knowledge base? Explain how, or why not.

Question 3: Clustering implementation + analysis

Implement K-means and K-medoid clustering on the given dataset and perform a comparative analysis. **Note that you cannot use any Python library for implementing the clustering methods.** However, you may use these standard libraries to implement and visualize: numpy, pandas, and matplotlib.

- Plot the elbow chart and identify the optimal value of K using your K-means implementation. The plot should show results for at least 10 SSE/WCSS values of K.
- Run the K-means and K-medoid algorithms on the given dataset using the optimal K. Your code must display the following results.
 - Iteration no.,
 - size of each cluster,
 - SSE/WCSS of each cluster, and
 - an overall SSE/WCSS value
- Display time complexity for both clustering methods in seconds.

Based on the above results, perform a comparative analysis of both clustering methods. Which clustering method has performed well in different contexts? Can you pick an overall winner for this dataset?

Question 4: Clustering library method + analysis

Implement both clustering methods from sklearn library using the optimal K used in the previous task. For comparison purposes, ensure that both clustering methods are run with the same settings as done in your own implementation, e.g., distance metric, stopping criterion, initial random points, etc. Show the following results in the end.

1. Total no. of iterations, size of each cluster, and an overall SSE/WCSS value.
2. Display time complexity for both clustering methods in seconds.

Based on the above results, perform a comparative analysis of sklearn's clustering methods and your implementation. How well your results align with the library's implementation? What could be the probable reasons for a change in results (if any).

Dataset Description (for Q# 3 & 4)

You are given a mines dataset in "Mine_Dataset.xlsx" file. A description of the attributes is given in "Information" sheet. The dataset to be used for clustering is given in "Normalized_Data" sheet. Perform clustering using the first three attributes/columns only (i.e., ignore the last column labeled 'M'). [After reading the dataset in a dataframe and having removed the last column, shuffle the records and then perform clustering.](#)